
Burn Depth Classification using Convolutional Neural Networks

Winter Term Project, Student name: Sofia Di Lucia

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DUE : *2nd February 2026*

PROJECT This assignment implies the following tasks:

Tasks: Describe the different tasks carried out in the project. Your final project should include:

- The Python or C++ source code and Jupyter sessions or code and data used to produce all figures and visual material.
- Slides for a 15 minutes seminar to present in the classroom the main results obtained in the project.
- The present document describing the project.

All the above mentioned project files should be in a gitlab project shared with the CNM professors where they can examine the project history.

PROBLEM DEFINITION

INTRODUCTION Correct assessment of burn depth is a fundamental step in the clinical management of burn injuries, as it directly affects treatment decisions and patient outcomes.

In particular, the distinction between superficial burns and deeper injuries (Fig.1) is crucial for determining whether surgical intervention, such as skin grafting, is required. However, accurate burn depth evaluation strongly depends on the visual experience of specialized clinicians and is therefore subject to variability and limited availability of expertise.

Automated analysis of burn images has the potential to reduce subjectivity, improve early treatment decisions and limit unnecessary patient transfers.

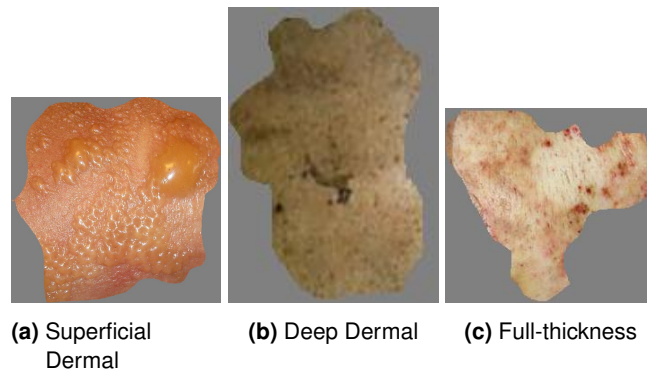


Figure 1: Example of the three different burn depth cases.

The present work is inspired by the study "*Segmentation and classification of burn images by color and texture information*" [1], in which color and texture descriptors extracted from segmented burn regions are used as inputs to a neural network classifier. While the original approach relies on handcrafted feature extraction and a Fuzzy-ARTMAP neural network, in this project we investigate burn depth classification using RGB images and convolutional neural networks.

CNNs allow to learn discriminative representations directly from resized color images, potentially reducing the need for manual feature design while preserving sensitivity to color and texture patterns. Due to the limited size of the available dataset, both binary and multiclass classification settings are explored.

COMPUTATION

DATASET The dataset used in this work consists of 94 RGB images of burn wounds and an excel file (`datainfo.xlsx`) containing the ground truth labels. A custom PyTorch Dataset class was implemented to manage image loading and label association. Each image is associated with a burn depth case, which belongs to one of the following three classes (Fig.1):

- Superficial dermal
- Deep dermal
- Full-thickness

The images are stored in a single dataset folder and are provided in two different formats (`.bmp` and `.jpg`). All images were converted to RGB format during data loading to ensure consistency.

PROBLEM FORMULATION Because of the different class distribution (Fig.2), two classification tasks were considered in this work:

1. binary classification was performed to distinguish between superficial dermal burns and deeper burns, which include deep dermal and full-thickness burns. This formulation is clinically motivated, as deeper burns often require similar treatment strategies and may require surgical intervention.
2. multi-class classification was performed to discriminate among superficial dermal, deep dermal and full-thickness burns. This task is more challenging due to the higher level of granularity and the limited number of samples, particularly for the full-thickness class.

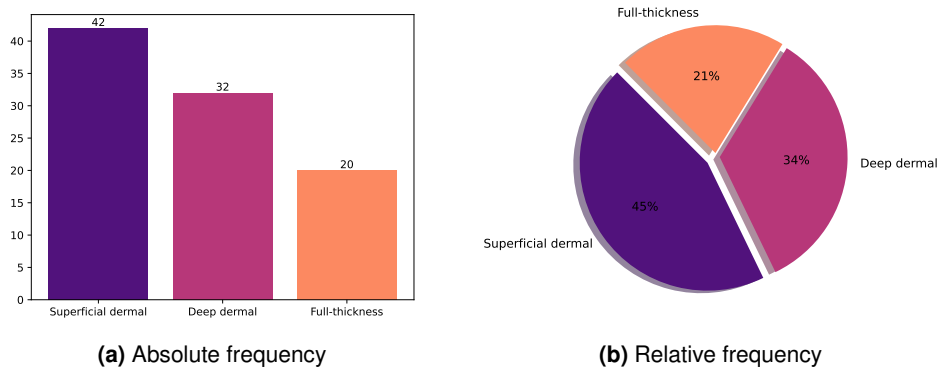


Figure 2: Class distribution of burn depth labels shows that the dataset is imbalanced. Superficial dermal burns are the most represented class (42 images, approximately 45%), while full-thickness burns are the least frequent (20 images, approximately 21%).

The class imbalance may bias the model toward the majority class, especially in the multi-class classification setting and represents an additional challenge given the limited size of the dataset.

DATA PREPROCESSING The images have different spatial resolutions (Fig.3) and irregular shapes.

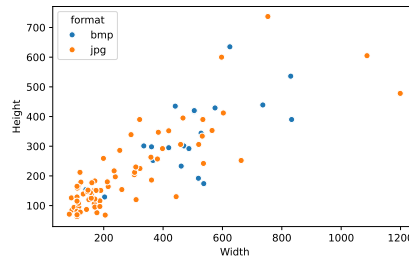


Figure 3: Size distribution of the dataset images: the sizes are distributed over a wide range, with a concentration of samples clustered around 200×200 pixels or below, indicating that a significant portion of the dataset consists of relatively small images.

In order to provide a consistent input to the neural network, all images were resized to a fixed resolution of 256×256 pixels and the pixel intensities were normalized to the range $[-1, 1]$ (Fig.4).

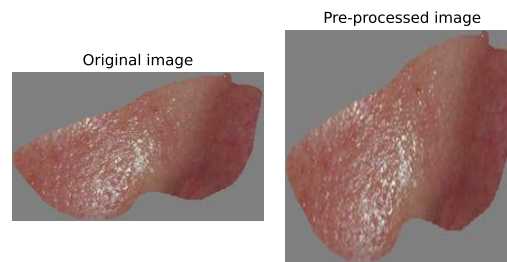


Figure 4: Example of preprocessing applied to a burn image.

Due to the limited size of the dataset, data augmentation techniques (random horizontal flips) were applied during training to increase variability while preserving the semantic content of the images.

Before training the model, categorical labels were encoded into numerical values. For the binary classification task, labels were encoded as follows:

- Class 0: Superficial dermal
- Class 1: Deep dermal and Full-thickness

For the multi-class classification task, labels were encoded as:

- Class 0: Superficial dermal
- Class 1: Deep dermal
- Class 2: Full-thickness

Two separate encoding functions were implemented to handle the different problem formulations.

NEURAL NETWORK ARCHITECTURE A custom convolutional neural network was designed to address the small size of the dataset and reduce the risk of overfitting. The architecture (Fig.5) consists of three convolutional blocks, each composed of a convolutional layer followed by a ReLU activation function and a pooling operation. The final convolutional block is followed by a global average pooling layer, which reduces the spatial dimensions and limits the number of trainable parameters. The extracted feature vector is then passed to a fully connected layer that produces the final class scores. The same architecture is used for both classification tasks, modifying only the number of output neurons.

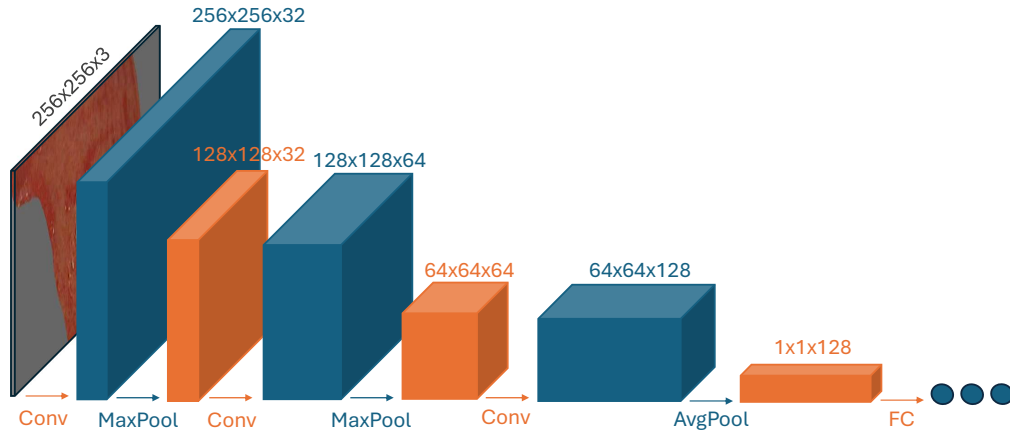


Figure 5: Custom convolutional neural network architecture.

TRAINING PROCEDURE The dataset was divided into training, validation, and test sets (70% – 15% – 15%) using a stratified split to preserve class proportions. Mini-batch training was adopted to improve optimization stability and computational efficiency. The model was trained using the Cross-Entropy loss function, which is appropriate for both binary and multi-class classification problems.

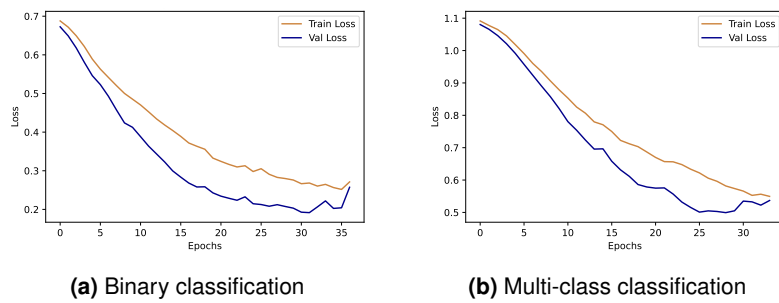


Figure 6: Training and validation loss during model training. In the binary case, losses decrease over epochs with validation loss lower than training loss, indicating stable convergence and no evident overfitting. In the multi-class case, losses also decrease but with slower convergence and higher values, reflecting the increased task complexity and the presence of class imbalance.

Optimization was performed using the Adam optimizer with weight decay for regularization. Early stopping based on the validation loss was employed to prevent overfitting.

(Fig.6) and to select the best-performing model. Training was performed for a maximum number of 100 epochs and the model parameters corresponding to the lowest validation loss were retained.

RESULTS

Model performance was evaluated on an independent test set not used during training or validation and accuracy was used as the primary evaluation metric. Additionally, confusion matrices were computed to analyze class-wise performance and to better understand common misclassification patterns, particularly in the multi-class scenario.

FIRST TASK The binary classification model achieved an overall accuracy of 86.7% on the test set.

As shown in the confusion matrix (Fig.7) and classification report (Tab.1), the model correctly classified all deep dermal and full-thickness burns, while superficial dermal burns were classified with a recall of 0.57, indicating that a portion of superficial cases was misclassified as deeper burns.

This behavior suggests that the model tends to overestimate burn severity rather than underestimate it.

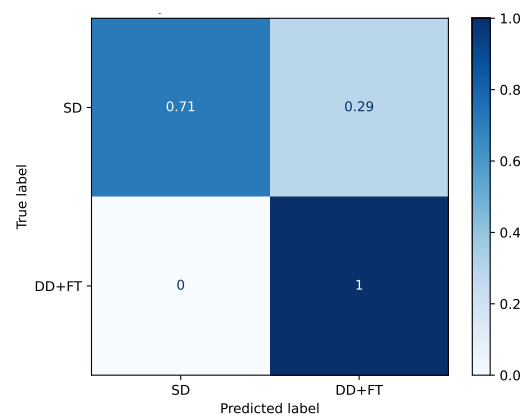


Figure 7: Confusion matrix for binary classification (SD = Superficial Dermal, DD+FT = Deep Dermal + Full-thickness).

Table 1: Classification report for the binary classification task on the test set.

Class	Precision	Recall	F1-score
Superficial Dermal (SD)	1.00	0.57	0.73
Deep Dermal + Full Thickness (DD+FT)	0.73	1.00	0.84

SECOND TASK For the multi-class classification task, the model achieved an overall accuracy of 66.7% on the test set. As shown in the classification report (Tab.2), performance varies significantly across classes.

Superficial dermal burns were classified with reasonable precision and recall, showing

a behavior consistent with the binary classification task. Deep dermal burns samples are correctly identified, although with lower precision due to confusion with neighboring classes. In contrast, the model failed to correctly classify full-thickness burns and this result is mainly attributed to the very limited number of full-thickness samples available for training and testing, combined with class imbalance and visual similarity between deep dermal and full-thickness burns.

Overall, these results highlight the increased difficulty of the multi-class problem and confirm that the binary classification task is more robust in low-data scenarios.

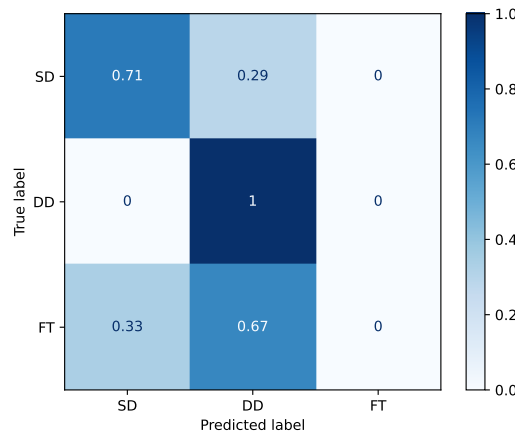


Figure 8: Confusion matrix for multi-class classification (SD = Superficial Dermal, DD = Deep Dermal, FT = Full-thickness).

Table 2: Classification report for the multi-class classification task on the test set.

Class	Precision	Recall	F1-score
Superficial Dermal (SD)	0.83	0.71	0.77
Deep Dermal (DD)	0.56	1.00	0.71
Full-thickness (FT)	0.00	0.00	0.00

CODE The source code, the dataset and the trained models are available at the following repository: github.com/sofiadilucia/cnm_project.git

REFERENCES

- [1] Begoña Acha Pinero, Carmen Serrano, Jose Ignacio Acha, and Laura M. Roa. Segmentation and classification of burn images by color and texture information. *Journal of Biomedical Optics*, 10(3):034014, 2005.